Session 52 — Diagnostic of prior knowledge

Revision and consolidation

11/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Identify what has been learnt.

· Classify the errors.

· Plan the remediation.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 25% increase

2. Give the formula for the general term of the arithmetic sequence from u_0. .....

3. Solve $10^{x} = 1000$. .....

4. Differentiate $x + \frac{4}{x}$. .....

5. State the law of total probability with two branches

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: identify what is known and classify the mistakes.

Investigation

Answer 14 questions, two per chapter, without notes. Record each error in a grid: knowledge, choice of method, calculation, reading, writing. Then choose two personal priorities.

Written summary

The diagnosis is used to choose the remediation: recognise the error, recall the relevant rule, redo a similar question, then check without help.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.

2. Solve an exponential equation with a logarithm.

3. Calculate a probability from a tree and then interpret.

References in the textbook

Extensions: Revisiting chapters 1 to 7. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the textbook.

Calao textbook · p. 29, Exercise 52

In France, since 2016, sales of hybrid vehicles have risen sharply. The table below gives the number of hybrid vehicles sold in France per year.

Year	2016	2017	2018
Number of hybrid vehicles sold	7 420	11 010	15 350

Part A

1. Determine the percentage change between 2016 and 2017, then between 2017 and 2018.
2. Determine the geometric mean of 1.48 and 1.39.
3. What is the average annual rate of change between 2016 and 2018?

Part B

Assume that from 2018, sales of hybrid cars increase by 43% per year and that this trend will continue. Sales are modelled by the sequence (u_n), where the integer part of u_n represents the number of hybrid cars sold during the year $(2018 + n)$.

1. What is u_0 ? Calculate u_1 and u_2.
2. What type of sequence is (u_n) ? Express u_n as a function of n.
3. According to this model, how many hybrid vehicles will be sold in 2023?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 380 and the rate or common ratio with 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 53 — Revisiting: exponential functions

Revision and consolidation

13/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Revisit exponential models.

· Solve and interpret.

· Check the units.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $3\quad x + 7 - x + 2$. .....

2. Solve $10^{\,x} = 1000$. .....

3. Differentiate $x + \dfrac{4}{x}$. .....

4. State the law of total probability with two branches

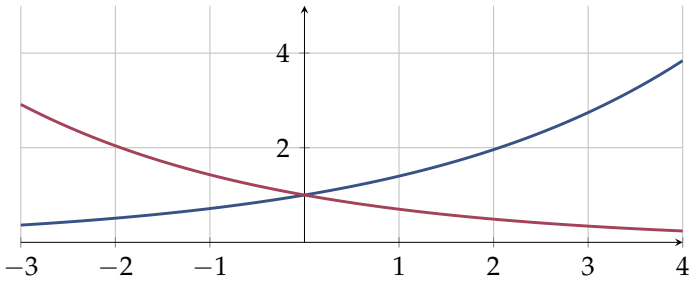
5. Give the formula for the general term of the arithmetic sequence from u_0. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: revisit exponential models and solve and interpret.

Investigation

A value of 18,000 increases by 3 ,5% per year. Write the model, calculate the value after eight years and determine when it will exceed 25,000.



Written summary

An exponential function models a constant relative change. First identify the initial value and the multiplier.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.

2. Solve an exponential equation with a logarithm.

3. Calculate a probability from a tree and then interpret.

References in the textbook

Extensions: Calao, p. 48, nos. 1 to 8, then p. 54, nos. 47 to 50. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 54, Exercise 58

The number of members of a basketball club has increased over the last three years by 2 ,5 %, then by 4.1% and finally by 3.8%. Calculate the average annual rate of change.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 385 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 54 — Revisiting: common logarithm

Revision and consolidation

18/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Revisit logarithms.

· Solve an equation.

· Check the conditions.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 5 per cent of 160 .....

2. Differentiate $x + \frac{4}{x}$. .....

3. State the law of total probability with two branches

4. Give the formula for the general term of the arithmetic sequence from u_0. .....

5. Solve $10^x = 1000$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: revisit logarithms and solve an equation.

Investigation

Solve $10^x = 340$, $1{,}06^n = 2$ and $1{,}06^n \geq 2$. For each answer, distinguish the exact value, the approx. value and smallest integer index.

Written summary

The logarithm turns an exponential equation into an equation in the exponent. Conditions: base positive and different from 1, right-hand side positive.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.

2. Solve an exponential equation with a logarithm.

3. Calculate a probability from a tree and then interpret.

References in the textbook

Extensions: Calao, p. 70, nos. 1 to 10, then pp. 73-75, nos. 30 to 42. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 76, Exercise 43

Let x a strictly positive real number. The aim of the exercise is to determine an order of magnitude of x.

1. $x = 23.54$ and $y = 569{,}47$.

   a. The order of magnitude of a number is the power of 10 closest to that number. Give the order of magnitude of x and y.

   b. Using the calculator, determine $\log(x)$. Conjecture a link between $\log(x)$ and the order of magnitude of x. Check your conjecture with y.

2. $x = 5{,}4517$. Using a calculator, find the order of magnitude of x and $\log(x)$. Is the conjecture made in question 1 true?

3. Suppose that x is written in scientific notation: $x = a \times 10^n$ for $a \in [1;10[$ and n is an integer. Give the order of magnitude of x depending on the value of a.

4. Deduce a method linking the order of magnitude of a strictly positive real number and its common logarithm.

5. Deduce the order of magnitude of the following two numbers: 2.26^{102} and 365.256^{2019}.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 390 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 55 — **Revisiting: reciprocal function**

Revision and consolidation — 20/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Revisit the reciprocal function.

· Carry out a study.

· Draw and interpret.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 10% increase

2. State the law of total probability with two branches

3. Give the formula for the general term of the arithmetic sequence from u_0. .....

4. Solve $10^x = 1000$. .....

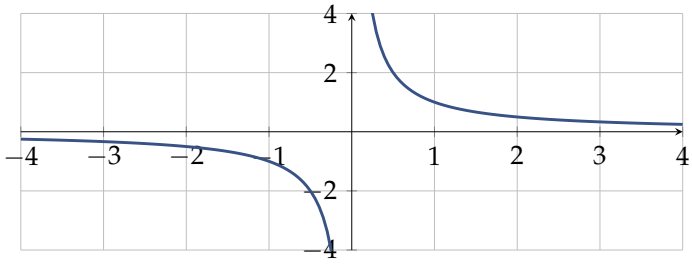
5. Differentiate $x + \dfrac{4}{x}$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: revisit the reciprocal function and carry out a study.

Investigation

Study $f(x) = x + \dfrac{16}{x}$ on $]0, +\infty[$ and interpret its minimum in a context of average cost.
Produce the variation table and a sketch.



Written summary

For a function containing $\dfrac{1}{x}$, first exclude $x = 0$, then differentiate, factorise and study the sign before the variations.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

References in the textbook

Extensions: Calao, p. 94, nos. 1 to 9, then p. 96, nos. 10 to 21. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 102, Exercise 47

In Murano, a glassmaker calculated that the cost of production (in euros) of x vases is given by: $C(x) = x^2 + 30x + 400$. The craftsman wants to make as many vases as possible at the lowest possible average unit cost.

Part A

1. Calculate $C(0)$ and deduce the craftsman's fixed costs. What is the production cost of 20 vases? For 20 vases, calculate the average cost of producing one vase.
2. Let $f(x)$ the average unit cost for x vases made. For a production of at least one vase ($x \geqslant 1$), express $f(x)$ as a function of x.
3. We assume that the average unit cost can be expressed as $f(x) = x + 30 + \frac{400}{x}$. Calculate $f'(x)$. Show that $f'(x) = \frac{(x-20)(x+20)}{x^2}$. Study the sign of $f'(x)$ and deduce the variations of f. How many vases must the craftsman produce for the average cost per unit to be minimal? Give that cost.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 395 and the rate or common ratio with 3. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.